

Omar Amir

AI Engineer | LLM & RAG Engineer | NLP Engineer

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SUMMARY

AI / LLM Engineer focused on building production-ready retrieval-augmented generation and agentic systems. Hands-on experience designing end-to-end RAG pipelines, stateful multi-agent workflows with LangGraph, vector search, and clean FastAPI service layers with async MongoDB and Pydantic v2. Comfortable owning a system from ingestion and retrieval design through evaluation, escalation logic, and deployment. Strong Arabic and English NLP background (IELTS 8.0) with a peer-reviewed publication, plus experience mentoring 100+ students as a Teaching Assistant.

TECHNICAL SKILLS

- **LLM & GenAI:** RAG pipelines, agentic workflows, LangChain, LangGraph, prompt engineering, long-term memory (Trustcall), human-in-the-loop, confidence-based escalation, LLM evaluation
- **Vector Search & NLP:** ChromaDB, Qdrant, Pinecone, embeddings, semantic chunking, document ingestion, Hugging Face, NLTK, Arabic & English NLP
- **Models & Serving:** Groq (Llama 3.3 70B), OpenAI-compatible providers, Stable Diffusion fine-tuning, Gradio
- **Backend & MLOps:** FastAPI, Motor (async MongoDB), MongoDB, Pydantic v2, REST APIs, Docker, uv, Git/GitHub, lifespan context managers, clean architecture
- **Languages & Core:** Python, NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow

PROJECTS

Mizan – Egyptian Labor-Law Agentic RAG Assistant — *LangGraph, ChromaDB, Groq (Llama 3.3 70B), Trustcall, Gradio, HuggingFace Spaces*

Live: <https://huggingface.co/spaces/OmarAmir2001/mizan>

- Built an agentic RAG assistant answering Egyptian labor-law questions, orchestrated as a stateful graph in LangGraph with tool-calling and conditional routing.
- Grounded responses in a ChromaDB vector store over labor-law documents, using Groq-hosted Llama 3.3 70B for low-latency generation.
- Added long-term, per-user memory with Trustcall so the assistant retains context across sessions; deployed as an interactive Gradio app on HuggingFace Spaces.

Customer Support Agent – Institute Help Desk — *FastAPI, Motor, MongoDB, Qdrant, Pydantic v2, Docker*

GitHub: https://github.com/OmarAmir2001/Customer_Support

- Designed a production-style customer-support RAG service with a clean FastAPI layer (BaseController/DataController separation, lifespan-managed resources, async Motor + MongoDB).
- Implemented a full document-ingestion pipeline (chunking, embedding, indexing) over Qdrant for semantic retrieval, with MongoDB indexing for fast metadata lookups.
- Engineered confidence-based escalation with human-in-the-loop interrupts and long-term student memory, plus modular routers for chat, history, profile, escalation, and admin.

GitHub Repository Q&A Agent — *LangChain, Docker, REST API*

GitHub: <https://github.com/OmarAmir2001/GitHub-Repository-Q-A-Agent>

- Built an end-to-end LLM pipeline (ingestion, retrieval, deployment) exposed via a REST API with sub-200ms responses; containerized with Docker for reproducibility.

Visioneer – Graduation Project — *Stable Diffusion v1.5, PyTorch, Android*

GitHub: https://github.com/OmarAmir2001/VISIONEER_APP

- Fine-tuned Stable Diffusion v1.5 on 12,000+ floor-plan samples and delivered an AI-powered Android app for automated design generation.

Reservation Cancellation Prediction — *Scikit-learn, Feature Engineering*

GitHub: <https://github.com/OmarAmir2001/reservation-cancellation-prediction>

- Built a supervised model on 10,000+ records reaching 85% accuracy, using feature engineering and hyperparameter tuning to improve F1 by 12%.

EXPERIENCE

Teaching Assistant – Faculty of Computer Science *Culture and Science City, Egypt* 2025 – Present

- Mentored 100+ students in Machine Learning, Algorithms, and Data Structures, raising assignment completion rates ~20% through hands-on projects.
- Guided students building ML models in Python, TensorFlow, PyTorch, and Scikit-learn across preprocessing, training, and evaluation.

EDUCATION

B.Sc. Computer Science – Excellence with Honors *Culture and Science City, Egypt* 2020 – 2024

PUBLICATION & LANGUAGES

- **Publication:** [MGV Journal 2025 – doi.org/10.22630/MGV.2025.34.3.4](https://doi.org/10.22630/MGV.2025.34.3.4)
- **Languages:** English – Full Professional (IELTS 8.0), Arabic – Native, German – Professional Working Proficiency

SELECTED COURSES

Deep Learning and Machine Learning Specializations (DeepLearning.AI / Stanford), Mathematics for ML and Data Science, Build Chat Applications with OpenAI and LangChain, Vector Databases with Pinecone, Applied Text Mining in Python, LLM Zoomcamp (DataTalksClub)